# Code map and complete implementation references

Group 15 • Final assessment preparation • 5 November 2026

Use this map to locate the mechanism behind an exam answer. Each public file in the source manifest has a purpose below and its complete contents in the current code appendix. The inventory includes existing helpers and does not assign authorship to the group.

## Trace one task through the code

`codec.decode_event` -> `scheduler.dispatch_partition` -> `executor.prepare_commit` -> `engine.ExperimentStateManager.apply` -> Data commit transaction -> scheduler Kafka commit. Then Data claim -> `notification.NotificationHandler.deliver` -> Data lease result. For a read, Caddy -> Data metadata/version -> bounded snapshot or SQL/Redis -> ordered JSON.

## Changes and their provenance

The original implementation report includes the earlier code and complete original patch. The November report with code includes every incremental application, runner, test and deployment change from that session, with before/after hashes. Its baseline includes pre-existing uncommitted work; git HEAD alone is not the full development baseline.

This assessment-pack work changes documentation and its build/verification helpers. It does not alter the application, instructor Rust logic, test thresholds or live deployment. Read the performance chapter for which changes have isolated regressions, before/after observations, or only mechanism-level explanations.

## root configuration

| File | Purpose and reason |
|---|---|
| .dockerignore | Exclude local secrets, caches and irrelevant artifacts from the application build context. |
| .gitignore | Keep private runtime configuration, credentials and generated test state out of version control. |
| Makefile | Documented operator/test entry points, including variable tests and generator-only stopping. |
| compose.course-test.yml | Original instructor binaries, local truth/receiver, variable-generator settings and local panel wiring. |
| compose.demo.yml | External topic/group, mTLS mounts, known hostname exception and course notification destination. |
| compose.local.yml | Independent local Kafka/sink wiring and bounded disposable broker retention. |
| compose.monitoring.yml | Operational Prometheus/Grafana/node exporter, provisioning and retention limits. |
| compose.yml | Core service boundaries, health/dependency wiring, durable volumes, limits and capped logs. |
| monitoring.env.example | Public tunable template, including allocated storage; real values live in an excluded private file. |
| pyproject.toml | Python package/dependency/tool/test-marker configuration. |
| requirements.lock | Pinned application dependency set for repeatable builds. |

# deploy

| File | Purpose and reason |
|---|---|
| deploy/Caddyfile | Public-only route forwarding and gateway behavior; internal mutations are not exposed. |
| deploy/CourseTools.Dockerfile | Build unmodified instructor Rust binaries for independent validation. |
| deploy/CourseTools.Dockerfile.dockerignore | Restrict the instructor-tool build context; avoid including private/generated state. |
| deploy/Dockerfile | Application and gateway build stages; repeatable Python runtime and OpenAPI artifacts. |
| deploy/alerts.yml | Nine validated rules covering service/jobs/input, individual deadlines, pool wait and disk pressure. |
| deploy/course/prometheus.yml | Scrape local instructor metrics and explicit run status. |
| deploy/course/provisioning/dashboards/course.yml | Provision local course dashboard location. |
| deploy/course/provisioning/datasources/course.yml | Provision local course Prometheus and truth PostgreSQL datasources. |
| deploy/course/schema.sql | Separate original-validator truth/notification tables; not the application authority. |
| deploy/docs.html | Public API documentation page served by the gateway. |
| deploy/grafana/dashboards/temperature.json | Operational dashboard, including pool, cache, disk, costs and individual deadline misses. |
| deploy/grafana/provisioning/dashboards/default.yml | Provision operational dashboard location. |
| deploy/grafana/provisioning/datasources/prometheus.yml | Provision operational Prometheus datasource. |
| deploy/prometheus.yml | Operational service/node scrape targets and alert rule loading. |

## loadtests

| File | Purpose and reason |
|---|---|
| loadtests/__init__.py | Package marker; no standalone service behavior. |
| loadtests/course.py | Original Rust self-test orchestration, ground-truth checks, variable phase evidence, panels and source/image build reuse. |
| loadtests/http_profile.py | Independent intended-send scheduling with bounded pending work, omitted-offer detection and phase drain accounting. |
| loadtests/matrix.py | Repeat deterministic profiles across executor/polling settings and save sizing evidence. |
| loadtests/notification_sink.py | Local independent-test receiver and fault injection; distinct from the original cryptographic validator. |
| loadtests/recovery.py | Exercise persisted partial groups, logical replay, delivery failures, component restarts and cold-cache SQL recovery. |
| loadtests/stress.py | Independent end-to-end workload driver, complete-history/alert validation, strict deadlines and phase/telemetry results. |
| loadtests/telemetry.py | Sample host CPU/disk, queue and Data operation counters; diagnostics include cohosted test tools. |
| loadtests/validation.py | Strict below-ten deadline predicate and conservative settled native-variable phase validation. |
| loadtests/workload.py | Build deterministic sensor/event truth, serialize supplied Avro and calculate observed distributions. |

## scripts

| File | Purpose and reason |
|---|---|
| scripts/build.sh | Existing course repository build helper; inclusion does not claim new authorship. |
| scripts/check-course-kafka.py | Read-only external mTLS preflight using a separate temporary consumer group; never publishes or commits live offsets. |
| scripts/generate-group-tokens.sh | Existing token-generation helper; generated credentials are excluded. |
| scripts/prepare-monitoring-credentials.py | Prepare the narrow mounted credential files without printing their private values. |
| scripts/push.sh | Existing course image-push helper; not executed to prepare this pack. |
| scripts/save_env.sh | Existing environment helper; runtime secret files are excluded from the report. |
| scripts/setup_env.sh | Existing development environment helper; inclusion does not claim new authorship. |

# temperature_observability

| File | Purpose and reason |
|---|---|
| temperature_observability/__init__.py | Package marker; no standalone service behavior. |
| temperature_observability/api.py | Development REST/OpenAPI contract; deployed Caddy routes implement the public path. |
| temperature_observability/autoscale.py | Opt-in bounded host policy with busy/idle hysteresis and dry-run; not continuously active in demo. |
| temperature_observability/codec.py | Avro object-container decoding and authoritative record_name header; returns explicit invalid envelopes. |
| temperature_observability/common.py | Shared metrics, asynchronous pooled transport, logging and supervised service tasks; transport tests guard compatibility. |
| temperature_observability/config.py | Validated service limits, Kafka/TLS options and one-VM plus provisioned-storage cost model. |
| temperature_observability/data.py | Authoritative atomic commit/load, version checks, query caches, writer fencing, outbox leases and diagnostic metrics. |
| temperature_observability/discovery.py | Discover executor addresses/load and reserve least-busy capacity without relying solely on stale observations. |
| temperature_observability/engine.py | Pure domain state machine: complete groups, ordered drain, phases, boundaries, deduplication and transition jobs. |
| temperature_observability/executor.py | Load snapshots, compute commit proposals, retry optimistic conflicts, enforce capacity; no authoritative worker-local state. |
| temperature_observability/healthcheck.py | Container probe entry point; readiness is distinct from business correctness under workload. |
| temperature_observability/history.py | Bounded versioned small-history snapshots, inclusive binary-search selection and interval rendering. |
| temperature_observability/models.py | Typed events, states, task lookups, commit proposals, partial groups, measurements and notification payloads. |
| temperature_observability/notification.py | Leased durable-job dispatch, token read, idempotency key, timeout/retry and individual deadline accounting. |
| temperature_observability/scheduler.py | Ordered partition dispatch, durable-before-Kafka acknowledgement, global/per-partition backpressure and status. |
| temperature_observability/schema.sql | Application tables/constraints and query/outbox indexes; PostgreSQL is the source of truth. |

## tests

| File | Purpose and reason |
|---|---|
| tests/test_codec.py | Avro decoding, termination header compatibility and invalid-message behavior. |
| tests/test_cost.py | Provisioned-storage billing estimate and incomplete-configuration behavior. |
| tests/test_course_build_cache.py | Source/image fingerprint invalidation, including schemas and SQL metadata while excluding credentials. |
| tests/test_data_integration.py | Real PostgreSQL atomicity/version rollback, leases, replay, cache coherence, delayed writers and large-history fallback. |
| tests/test_engine.py | Domain means, lifecycle, allocated sensors, incomplete groups, duplicates and excursion semantics. |
| tests/test_history.py | Inclusive bounds, duplicate timestamps, version rejection, interval closure and entry/point eviction limits. |
| tests/test_load_validation.py | Reject misleading aggregate/phase results, omitted offered work and exactly-ten/invalid latency observations. |
| tests/test_notification.py | Delivery outcomes, retries/lease acknowledgement and strict deadline miss accounting. |
| tests/test_public_api_integration.py | Required public routes and query contract, including internal-route protection. |
| tests/test_scheduler.py | Ordering, reservation/capacity, backpressure and reassignment behavior; scaling policy boundaries. |
| tests/test_service_client.py | Shared HTTP transport response/error compatibility and compressed responses. |

## Documentation tooling

prepare_sources.py verifies the source snapshot and generates this map and full code appendix. build_reports.py adapts the earlier report renderer, reads saved JSON, draws figures and builds PDFs. verify_reports.py checks numeric claims, local links, PDF content/bounds, source identity and private-value exclusion, then creates a manifest and portable ZIP. These tools never start workloads or change Docker services.

The 69-file public snapshot does not include every unchanged Rust source in the assignment. Those remain the instructor repository; the local runner compiles them unchanged. The portable pack is for reading and review, not a replacement for the full deployment checkout.